

Acute Kidney Injury

(Abrupt rise in serum Cr)



Key point: eGFR is not meaningful in setting of AKI. eGFR is based on assumption of stable GFR

Go To Cases

INTRA-renal

- 1. "Active" urine sediment (supports intra-renal)
- 2. Elevated FeNa while oliguric (rules out pre-renal)
- 3. Normal kidney US (rules out post-renal).

PRE-renal

Etiologies

- ↓ Effective Arterial Blood Volume (EABV) & ↓ Renal Vein Outflow

Dx

- 1. History
- 2. Bland urinalysis
- 3. No hydronephrosis
- 3. FeNa <1% (if oliguric)

↓ EABV

Renal artery

Renal vein

Glomerulus

Afferent arteriole

Efferent arteriole

Glomerular

Acute Glomerulonephritis (GN)

Dx: dysmorphic RBCs / RBC casts, proteinuria, leukocyturia, +/- HTN

Non-Crescentic & Crescentic ("RPGN")

Tubulointerstitial

Tubular

- 1. Acute tubular necrosis (ATN)*
- 2. Tubular obstruction Acute interstitial nephritis

Acute Interstitial

- 1. Drugs
- 2. Infection
- 3. Systemic immune dz

Vascular

Vasogenic

- 1. NSAIDs
- 2. Calcineurin inhibitors (CNIs)
- 3. ACE-I, ARBs

Microangiopathic Hemolytic Anemia (MAHA) (e.g. TTP/HUS)

Cholesterol Emboli

POST-renal

Etiology

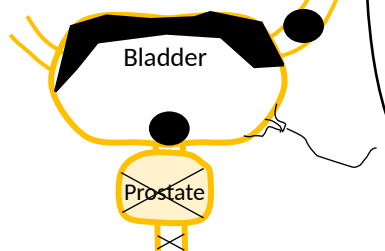
Obstructed outflow of **BOTH kidneys** or the **SINGLE FUNCTIONING kidney**

D

- 1. History
- 2. Hydronephrosis on ultrasound of kidney

Bladder

Prostate



Click

Case 1

35yo woman presenting to ED with one week of vomiting and diarrhea. Reports decreased UOP and light headedness

Cr **1.5** (baseline 0.7) | BUN **45** | FeNa <0.2%
Urinalysis: hyaline casts | Kidney US: Normal

Pre-renal azotemia from hypovolemia

Control nausea, aggressive rehydration

Case 2

70yo man presenting to clinic with difficulty initiating urination, weak stream and incomplete emptying for the past couple of years that has significantly worsening over the past week since he started taking allergy medications.

Cr **2.0** (baseline 0.8) | BUN **40** | FeNa 1.5%
Urinalysis: nml | Kidney US: Bilateral mod hydronephrosis

Post-renal nephropathy from BPH + medication induced urinary retention.

Discontinue anti-cholinergic, start tamsulosin, consider urinary catheter

Case 3

23yo transman with SLE not currently on therapy presents to the ED with arthralgias, dark urine, decreased urine output.

Cr **3.6** (baseline 0.9) | BUN 30 | FeNa 2.8%
Urinalysis: dysmorphic RBCs and RBC casts
Kidney US: Normal

Likely acute glomerular nephritis from lupus flare

Induction therapy with steroids + cyclophosphamide or mycophenolate

Case 4

40yo woman returns to clinic with decreased UOP and fatigue a 5 days into a 14 day course of amoxicillin/clavulanate for acute sinusitis. Noted to have a faint truncal maculopapular rash.

Cr **2.5** (baseline 0.9) | BUN 30 | FeNa 2.2%
Urinalysis: proteinuria, WBCs & WBC casts. No eosinophils
Kidney US: Normal

Likely acute interstitial nephritis (AIN) from beta-lactam

Discontinue abx, repeat labs in 3 days and consider bx +/- steroids if not improving

Case 5

42yo man with DMII is admitted for pneumonia with severe sepsis and hyperglycemia. He received aggressive IVF resuscitation but his Cr continues to rise 24hr after admission and he is now anuric.

Cr **3.4** (2.5 on admission, baseline 1.2) | FeNa 2.3%
Urinalysis: muddy brown casts
Kidney US: Normal

Likely ischemic acute tubular necrosis from prolonged hypoperfusion due to hypovolemia and sepsis

Monitor electrolytes, maintain euvoolemia, avoid overhydration (anuric), and avoid nephrotoxins.

Case 6

36yo man presents to the ED with diffuse muscle pain not responding to NSAIDs and dark urine two days after starting a new high intensity work-out program.

Cr **3.6** (baseline 1.2) | BUN 30 | FeNa 3% | CK 20,000
Urine dipstick: 3+ blood, no RBC on urinalysis
Kidney US: Normal

Likely tubular obstruction from myoglobin as a result of rhabdomyolysis PLUS injury 2/2 NSAIDs

Aggressive fluid resuscitation and discontinuation of NSAIDs

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Framework Notes (current)

The pre-/intra-/post-renal framework and the bedside tests on this board remain standard practice.

AKI definition (KDIGO 2012, unchanged): $\uparrow \text{Cr} \geq 0.3 \text{ mg/dL}$ within 48 h, OR $\uparrow \text{Cr} \geq 1.5 \times$ baseline within 7 days, OR urine output $< 0.5 \text{ mL/kg/h}$. eGFR is NOT valid during AKI (assumes steady state).

FeNa $< 1\%$ suggests pre-renal but only when oliguric and off diuretics; FeUrea $< 35\%$ is more reliable on diuretics. Always rule out obstruction early with renal US — it is fast and the lesion is reversible.

Management by category: pre-renal $\rightarrow$ restore perfusion; intra-renal $\rightarrow$ treat cause, remove nephrotoxins (NSAIDs/contrast/ACEi-ARB), supportive care; post-renal $\rightarrow$ relieve obstruction.

Take-Home Points

1. AKI = abrupt rise in Cr (KDIGO thresholds). eGFR is meaningless during AKI.
2. Three buckets: PRE-renal (↓ EABV), INTRA-renal (glomerular / tubulointerstitial / vascular), POST-renal (obstruction of both kidneys or a single functioning kidney).
3. Three quick discriminators: history + bland UA + FeNa <1% (if oliguric) → pre-renal; active sediment → intra-renal; hydronephrosis on US → post-renal.
4. Rule out obstruction EARLY with renal US — easily reversible. Unilateral obstruction does NOT cause AKI unless it is the single functioning kidney.
5. ATN is the most common intra-renal cause (ischemic or toxic); muddy-brown granular casts. Remove nephrotoxins and maintain euvolemia.

Attribution & Changelog

Adapted from the TeachIM original “Acute Kidney Injury” (Mack Holmberg, MD; expert review Raj Mehrotra, MD, MS, Univ. of Washington Nephrology). Published 2020. teachim.org.

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Corrections (surgical, animations preserved): “Glomerularnephritis” → “Glomerulonephritis” (presenter + learner board); “Inta-Abdominal” → “Intra-Abdominal” (learner board).

Added: framework-notes slide (KDIGO criteria, FeNa/FeUrea caveat) — standing teaching unchanged.

Companion files: presenter board + learner board (both corrected) + AKI learner handout (reviewed). Educational use only; clinician-review before teaching.